

AERO 689 Final Project Report Instructions

Aerospace Engineering, Texas A&M University

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1. Purpose of the Final Report

The preliminary project report was intended to define the problem, motivate the aerospace relevance, and give the instructor an opportunity to provide early feedback. The final report should now show how the project was actually completed.

Students should revise their preliminary report using the feedback they received and expand it into a complete technical report. The final submission should describe a working machine learning, data-driven, or hybrid physics/ML system; present results; analyze the quality of those results; and discuss what was learned from the study.

This is no longer a proposal-only document. By the final submission, the project should describe a working system, processed or simulated data, trained or evaluated models, and evidence that the system was tested.

2. Final Report Expectations

The final report should read like a concise technical paper for an aerospace AI or machine learning application. It should be professional, clearly organized, and supported by quantitative evidence.

Students should demonstrate:

- A clear aerospace engineering problem and motivation.
- A specific machine learning, statistical, optimization, or hybrid modeling approach.
- A working implementation of the proposed system.
- Data collection, generation, preprocessing, and validation steps.
- Results from experiments, simulations, or model evaluations.
- Discussion of model performance, failure cases, limitations, and engineering implications.
- Identification of existing tools, methods, or systems that address similar problems.
- Comparison with existing tools or a simple baseline when feasible. A baseline comparison is not required for every project, but it will strengthen the report when an appropriate baseline is available.
- Evidence that feedback from the preliminary report was incorporated.
- Proper citation of datasets, references, tools, software, and any AI assistance.

The final report should generally be about **6-10 pages**, excluding references and optional appendices. Shorter reports are acceptable only if they are complete and technically dense. Longer reports should avoid unnecessary background material and focus on the project work, results, and analysis.

3. Required Sections

1. Title, Team, and Contributions

Include the project title, team members, and each member's role or contribution. If the project was completed individually, state that clearly.

2. Abstract

Provide a short summary of the problem, approach, data, main results, and aerospace significance. The abstract should make clear what was built and what was found.

3. Introduction and Problem Statement

State the aerospace problem being addressed. Explain why the problem matters for aerospace engineering, operations, design, safety, autonomy, propulsion, structures, flight systems, or another relevant area.

This section should be revised from the preliminary report. It should be sharper and more specific now that the project has been completed.

4. Background, Related Work, and Existing Tools

Summarize the technical context needed to understand the project. Include relevant prior work, datasets, models, engineering methods, existing software tools, commercial systems, open-source packages, or aerospace applications.

Students should identify what tools or methods already exist for this problem or for a closely related problem. Explain what those tools do, what their limitations are, and how the project differs from or improves on them.

This section should not be a broad textbook survey. It should focus on the ideas and existing capabilities directly needed for the project.

5. Response to Preliminary Feedback

Briefly explain how feedback from the preliminary report was incorporated. This may include changes to the problem scope, dataset, model, evaluation metrics, figures, writing, or technical assumptions.

If a piece of feedback was not incorporated, explain why. The goal is to show that the preliminary review was used to improve the final work.

6. Data and Preprocessing

Describe the data used in the project. This may include experimental data, public datasets, simulation data, synthetic data, CFD data, flight data, structural sensor data, images, time-series data, or other sources.

Include:

- Data source and citation.
- Input variables and output targets or labels.
- Number of samples, trajectories, images, time steps, or cases.
- Train/validation/test split or cross-validation strategy.
- Preprocessing steps such as filtering, normalization, feature extraction, dimensionality reduction, augmentation, missing-data handling, or label construction.
- Any assumptions or limitations in the dataset.

If the dataset was generated by simulation, describe the simulation model and key parameters.

7. Methods and System Implementation

Describe the model, algorithm, or hybrid system that was implemented. The report should be detailed enough that another student in the course could understand the workflow and reproduce the main idea from the report.

Include:

- Model architecture or algorithm.
- Input and output definitions.
- Loss function, objective function, or optimization criterion.
- Training procedure and important hyperparameters.
- Software tools and libraries used.
- How the implemented system works as an end-to-end pipeline.
- How the system relates to existing tools or methods identified in the background section.

For AI-related projects, this section should make clear what was actually built. For example, specify whether the system performs classification, regression, anomaly detection, clustering, forecasting, control, surrogate modeling, optimization, image analysis, or physics-informed learning.

8. Experimental Setup and Evaluation Metrics

Explain how the system was evaluated. The evaluation should be aligned with the original aerospace problem.

Possible metrics include accuracy, precision, recall, F1 score, confusion matrix, ROC/AUC, RMSE, MAE, relative error, calibration error, uncertainty estimates, computational cost, inference time, physical constraint violation, trajectory error, or engineering performance metrics.

Include baseline comparisons when appropriate. A baseline may be a simple linear model, nearest-neighbor method, physics-only estimate, rule-based method, previous homework method, existing software tool, published result, or a simplified version of the proposed model.

If a direct baseline is not available, explain why and provide another form of comparison, such as qualitative comparison to existing tools, comparison to expected physical behavior, comparison across model variants, or comparison against a simple engineering estimate.

9. Results

Present the main quantitative and qualitative results. Use figures and tables where they help the reader understand the outcome.

Common useful results include:

- Training and validation curves.
- Test-set performance metrics.
- Confusion matrices or classification reports.
- Predicted vs. true plots.
- Error distributions.
- Comparison with an existing tool, existing method, or simple baseline when available.
- Example successes and failures.
- Ablation studies or parameter comparisons.
- Visualizations of learned features, clusters, latent spaces, trajectories, images, or physical fields.

Every figure and table should be referenced in the text and should have a clear caption. Do not include figures that are not discussed.

10. Discussion and Engineering Interpretation

Interpret the results. This is one of the most important parts of the final report.

Discuss:

- Whether the system solved the intended problem.
- What the results mean in aerospace engineering terms.
- Where the model performs well.
- Where the model fails or is unreliable.
- Whether the results are physically reasonable.
- Sensitivity to data quality, model assumptions, hyperparameters, or operating conditions.
- Tradeoffs between model complexity, accuracy, interpretability, and computational cost.
- How the system compares with existing tools, prior methods, or a baseline.

The discussion should go beyond reporting numbers. Explain why the results matter.

11. Limitations, Risks, and Future Work

Identify the main limitations of the current work. These may involve dataset size, label quality, simulation fidelity, generalization, uncertainty, overfitting, lack of real flight data, limited validation cases, computational cost, or missing physical constraints.

Then describe realistic next steps that would improve the system if more time were available.

12. Aerospace Impact

Explain how the completed project connects back to aerospace engineering. Discuss how the method could support design, analysis, monitoring, control, autonomy, safety, operations, maintenance, or decision making.

This section should be specific to the project. Avoid generic claims that machine learning is useful.

13. Conclusion

Summarize what was built, the most important result, and the main technical takeaway. The conclusion should be short and direct.

14. References

Cite all datasets, papers, software packages, textbooks, web resources, and tools used in the project. References should be complete enough that the source can be found.

15. Appendix or Supplementary Material

Optional appendices may include extra derivations, additional plots, extended tables, or implementation details. The main report should still be understandable without the appendix.

4. Submission Format

Students should submit **one PDF file only**. Do not submit source code, notebooks, datasets, archives, or separate figure files unless the instructor specifically requests them later.

The PDF must contain all information needed to evaluate the project:

- The problem statement and motivation.
- The data source and preprocessing description.
- The implemented method or system.
- The experimental setup and evaluation metrics.
- Results, figures, tables, and discussion.
- References, including datasets, software tools, and existing methods.
- A short acknowledgement of AI tool use, if applicable.

Students should not include long code listings in the report. Short pseudocode, equations, pipeline diagrams, or concise implementation descriptions are acceptable when they clarify the method.

5. Figures and Tables

Good final reports usually contain several well-designed figures or tables. Examples include:

- A system pipeline diagram.
- A data summary table.
- A model architecture diagram.
- Training and validation curves.
- Main performance comparison table.
- Comparison with an existing tool, existing method, or simple baseline.
- Predicted vs. true result plot.
- Error or residual analysis.
- Representative examples of correct and incorrect model behavior.

Figures should be readable, labeled, and discussed in the text. Axes should have units where applicable. Tables should include clear column names and meaningful precision.

6. AI Tool Use and Academic Integrity

Students may use AI tools to help brainstorm, edit writing, improve implementation, or debug. However, students are responsible for the technical correctness of the final report and all results reported in it.

If AI tools were used, include a short acknowledgement describing how they were used. Do not submit AI-generated technical claims, citations, equations, or results without checking them carefully.

All datasets, external software tools, figures, and prior work must be properly cited. Do not present another person's work as your own.

7. Grading Rubric

The final report will be graded out of **100 points** using the criteria below.

Category	Points	What will be evaluated
Problem clarity and aerospace relevance	15	The problem is clearly defined, technically meaningful, and connected to aerospace engineering. The motivation, engineering consequences, and intended use case are specific and credible.
Incorporation of preliminary feedback	10	The report explains how feedback from the preliminary submission was addressed. Changes in scope, method, data, evaluation, figures, or writing are clearly described.

Category	Points	What will be evaluated
Background, related work, and existing tools	10	The report identifies relevant prior work and existing tools, methods, software packages, or systems that address similar problems. Their capabilities and limitations are discussed.
Technical approach and implementation	15	The selected ML, statistical, optimization, or hybrid method is appropriate for the problem. The implemented system is clearly described, technically coherent, and shown to be working. Important modeling choices and assumptions are justified.
Data and preprocessing	10	The data source, variables, labels or targets, sample size, train/validation/test strategy, preprocessing steps, and dataset limitations are clearly described.
Experimental setup, evaluation, and comparison	15	Evaluation metrics are appropriate for the aerospace problem. Experiments are technically meaningful. Comparison with an existing tool, prior method, engineering estimate, or simple baseline is included when feasible. If no baseline is available, the report explains why and provides another reasonable comparison.

Category	Points	What will be evaluated
Results, discussion, and engineering interpretation	15	Results are presented with clear figures, tables, and quantitative metrics. The discussion explains performance, limitations, failure cases, physical reasonableness, and engineering meaning. Claims are supported by evidence.
Writing, organization, and professionalism	10	The report is concise, readable, well organized, and technically precise. Figures and tables are labeled and discussed. References, datasets, tools, and AI assistance are acknowledged where appropriate.

Strong reports will not only show that a model was trained, but will also explain whether the resulting system is useful for the aerospace problem, how it compares to alternatives, and where it should or should not be trusted.

8. Submission Checklist

Before submitting, make sure the final report includes:

- A clear statement of the aerospace problem.
- A summary of how preliminary feedback was addressed.
- Description of the dataset and preprocessing.
- Description of the implemented method or system.
- Identification of existing tools or methods for similar problems.
- Comparison with an existing tool, method, or simple baseline when feasible.
- Evidence that the system runs and produces results.
- Quantitative evaluation metrics.
- Figures or tables showing the main results.
- Discussion of what the results mean.
- Limitations and future work.
- Aerospace impact.

- References.
- A single PDF file for submission.
- Acknowledgement of AI tool use, if applicable.

The final submission should make it clear what was proposed, what was built, what results were obtained, and why those results matter for aerospace engineering.